

COURSEWORK SPECIFICATION

COMM040 - Text Mining and Natural Language Processing

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Machine Translation

Compare MT systems using overlap-based and neural metrics

Use Case

LinguaTech Solutions is a European translation provider working with public institutions and legal clients. The company wants to integrate Machine Translation (MT) to reduce turnaround time and cost, while maintaining high adequacy and fluency. Their main language pairs are:

- English → French
- English → German

They must choose between 2 MT paradigms (neural, instruction-tuned LLM) and base their decision on automatic evaluation, qualitative inspection, and cross-language robustness.

You are hired to conduct this evaluation and recommend a system.

Systems Under Evaluation

System A: Neural MT

- Model: Helsinki-NLP/opus-mt-en-fr
- Model: Helsinki-NLP/opus-mt-en-de
- Architecture: Transformer encoder–decoder
- Training: OPUS parallel corpora
- Strength: fluency and reordering

System B: Instruction-Tuned LLM

- Model: google/flan-t5-small
- Architecture: Transformer (encoder–decoder)
- Training: Large-scale pretraining + instruction tuning
- Strength: semantic flexibility
- Risk: paraphrasing and hallucination

Dataset

- Use wmt14 via datasets library.
- Use a subset (e.g., 200 sentences).
- Ensure access to multiple references per sentence.

Required Libraries

- datasets
- transformers
- sacrebleu
- bert-score
- numpy
- pandas

Task 1: Data Preparation

1. Load WMT14 test subset for EN→FR and EN→DE.
2. Extract:
 - a. Source sentences
 - b. All reference translations
3. Store references in a list-of-lists format (required for multi-reference evaluation).
4. Inspect the number of references per sentence.

Task 2: Generate Translations

For each system and language pair:

1. Load the model using `transformers.pipeline`.
2. Design a generation loop over the 200 sentences.
3. Store outputs in a structured dictionary:
outputs[system][language_pair]
4. Use deterministic decoding (no sampling).

Task 3: Automatic Evaluation

Compute the following metrics: BLEU, chrF, TER, BERTScore

1. Use `sacrebleu` for BLEU, chrF and TER.
2. Use `bert-score` for semantic similarity.
3. Ensure multi-reference scoring is correctly passed.
4. Compute results per system and per language pair.
5. Present results in a comparison table.

Task 4: Cross-System Comparison

1. Which system performs best under BLEU?
2. Does ranking change under chrF?
3. Does BERTScore favour a different system?
4. Are results consistent between FR and DE?